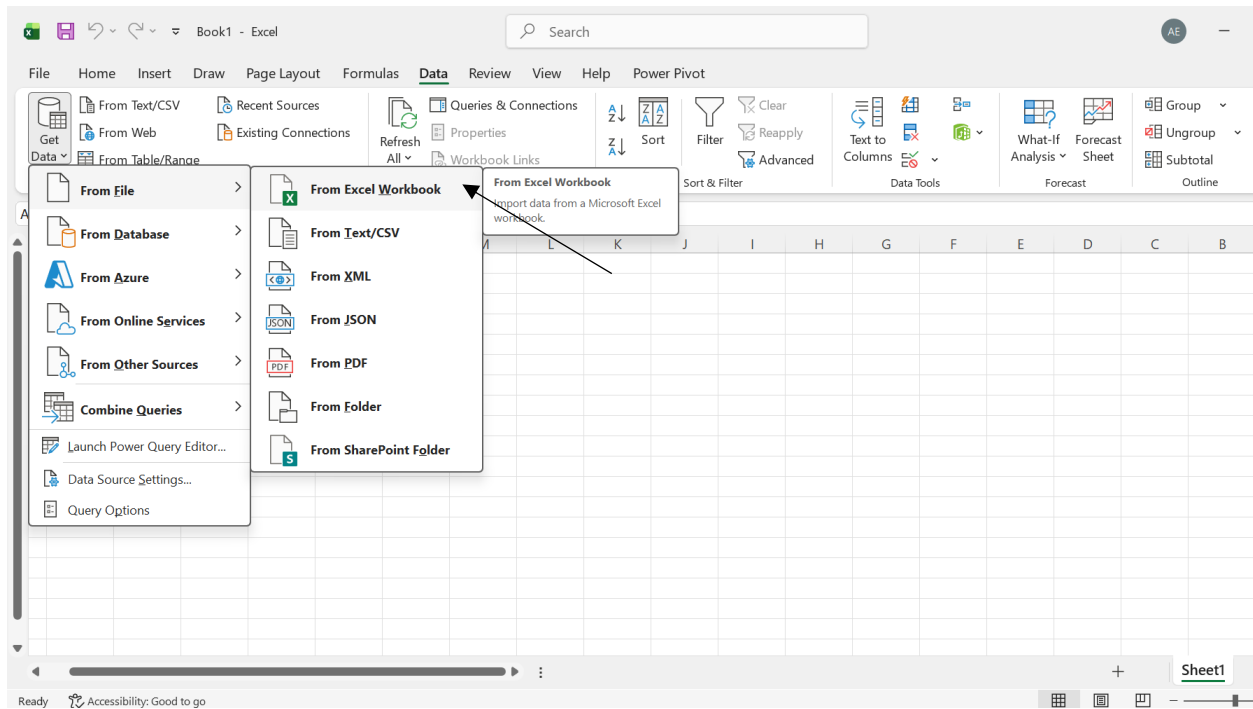
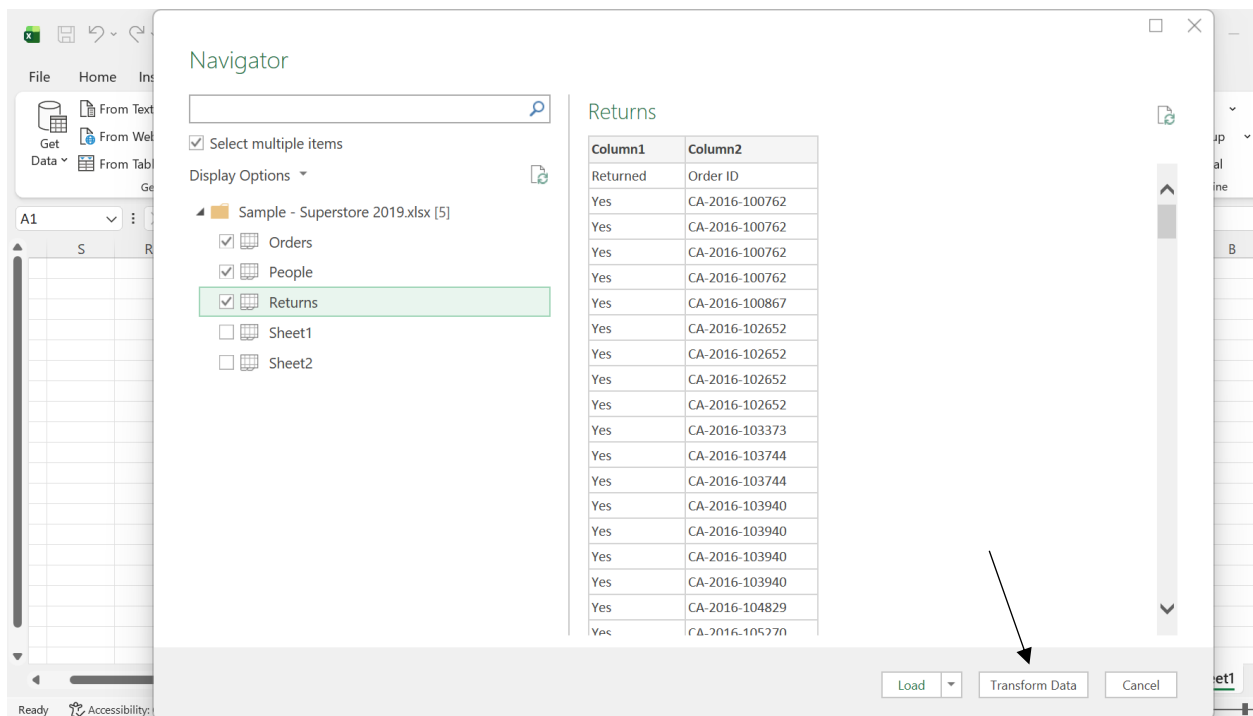


1.lunch power query editor:



2. choose excel file, import and transform:



3. choose column profiling based on entire data:

The screenshot shows the Power Query Editor interface. The 'Row ID' column is selected, and its data type is 'Int64.Type'. The 'Column profiling based on entire data set' option is selected in the bottom status bar. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list with 'Changed Type' selected.

Row ID	Order ID	Order Date	Ship Date
1	CA-2018-152156	11/8/2018	11/11/2018
2	CA-2018-152156	11/8/2018	11/11/2018
3	CA-2018-138688	6/12/2018	6/16/2018
4	US-2017-108966	10/11/2017	10/18/2017
5	US-2017-108966	10/11/2017	10/18/2017
6	CA-2016-115812	6/9/2016	6/14/2016
7	CA-2016-115812	6/9/2016	6/14/2016
8	CA-2016-115812	6/9/2016	6/14/2016
9	CA-2016-115812	6/9/2016	6/14/2016
10	CA-2016-115812	6/9/2016	6/14/2016
11	CA-2016-115812	6/9/2016	6/14/2016
12	CA-2016-115812	6/9/2016	6/14/2016
13	CA-2019-114412	4/15/2019	4/20/2019
14	CA-2018-161389	12/5/2018	12/10/2018
15	US-2017-118082	11/22/2017	11/26/2017

4. make row ID text:

The screenshot shows the Power Query Editor interface. The 'Row ID' column is selected, and its data type is 'Text'. The 'Column profiling based on entire data set' option is selected in the bottom status bar. The 'Query Settings' pane on the right shows the 'APPLIED STEPS' list with 'Changed Type' selected.

Row ID	Order ID	Order Date	Ship Date
1	CA-2018-152156	11/8/2018	11/11/2018
2	CA-2018-152156	11/8/2018	11/11/2018
3	CA-2018-138688	6/12/2018	6/16/2018
4	US-2017-108966	10/11/2017	10/18/2017
5	US-2017-108966	10/11/2017	10/18/2017
6	CA-2016-115812	6/9/2016	6/14/2016
7	CA-2016-115812	6/9/2016	6/14/2016
8	CA-2016-115812	6/9/2016	6/14/2016
9	CA-2016-115812	6/9/2016	6/14/2016
10	CA-2016-115812	6/9/2016	6/14/2016
11	CA-2016-115812	6/9/2016	6/14/2016
12	CA-2016-115812	6/9/2016	6/14/2016
13	CA-2019-114412	4/15/2019	4/20/2019
14	CA-2018-161389	12/5/2018	12/10/2018
15	US-2017-118082	11/22/2017	11/26/2017

5. make sure if data type of columns is true:

Table.TransformColumnTypes(#"Changed Type",{{"Row ID", type text}})

Order Date	Ship Date	Ship Mode
11/8/2018	11/8/2018	Second Class
6/12/2018	6/12/2018	Second Class
10/11/2017	10/11/2017	Standard Class
6/9/2016	6/9/2016	Standard Class
6/9/2016	6/9/2016	Standard Class
6/9/2016	6/9/2016	Standard Class
6/9/2016	6/9/2016	Standard Class
6/9/2016	6/14/2016	Standard Class
6/9/2016	6/14/2016	Standard Class
6/9/2016	6/14/2016	Standard Class
4/15/2019	4/20/2019	Standard Class
12/5/2018	12/10/2018	Standard Class
11/11/2017	11/16/2017	Standard Class

TransformColumnTypes(#"Changed Type",{{"Row ID", type text}, {"Sales", type decimal}, {"Quantity", type decimal}}

Sales	Quantity
114.9	1706.184
911.424	15.552
407.976	60.81

Query Settings

PROPERTIES

Name: Orders

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Changed Type1

6. remove duplicates:

Remove Duplicates

Row ID	Order Date	Ship Date
1	11/8/2018	11/11/2017
2	11/8/2018	11/11/2017
3	6/12/2018	6/16/2018
4	10/11/2017	10/18/2017
5	10/11/2017	10/18/2017
6	6/9/2016	6/14/2016
7	6/9/2016	6/14/2016
8	6/9/2016	6/14/2016
9	6/9/2016	6/14/2016
10	6/9/2016	6/14/2016
11	6/9/2016	6/14/2016
12	6/9/2016	6/14/2016

Query Settings

PROPERTIES

Name: Orders

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Changed Type1

7. standardize category names:

The screenshot shows the Power Query Editor interface. The 'Format' menu is open, and the 'Trim' option is selected. A tooltip explains: "Remove leading and trailing whitespaces from each cell in the selected columns." The data table shows columns for Product ID and Sub-Category.

Product ID	Sub-Category
FUR-BO-10001798	Furniture
FUR-CH-10000454	Furniture
OFF-LA-10000240	Office Supplies
FUR-TA-10000577	Furniture
OFF-ST-10000760	Office Supplies
FUR-FU-10001487	Furniture
OFF-AR-10002833	Office Supplies
TEC-PH-10002275	Technology
OFF-BI-10003910	Office Supplies
OFF-AP-10002892	Office Supplies
FUR-TA-10001539	Furniture
TEC-PH-10002275	Technology

8.remove blanks:

The screenshot shows the Power Query Editor interface. The 'Remove Rows' menu is open, and the 'Remove Blank Rows' option is selected. A tooltip explains: "Remove all blank rows from this table." The data table shows columns for Sales, Discount, and Profit.

Sales	Discount	Profit
261.96	0	41.9136
731.94	0	219.582
14.62	0	6.8714
957.5775	0.45	-383.031
22.368	0.2	2.5164
48.86	0	14.1694
7.28	0	1.9656
907.152	0.2	90.7152
18.504	0.2	5.7825
114.9	0	34.47
1706.184	0.2	85.3092
911.424	0.2	68.3568
15.552	0.2	5.4432
407.976	0.2	132.5922
68.81	0.2	172.858

9. select nulls only to understand and handle it:

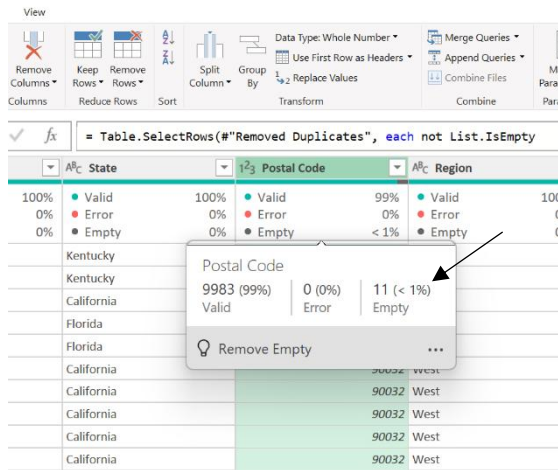


Table: SelectRows("#Removed Duplicates", each not List.IsEmpty)

State	Postal Code	Region
Kentucky		
Kentucky		
California		
Florida		
Florida		
California		
California		
California		
California		
California		

Postal Code statistics: 9983 (99%) Valid, 0 (0%) Error, 11 (< 1%) Empty. Remove Empty button highlighted.

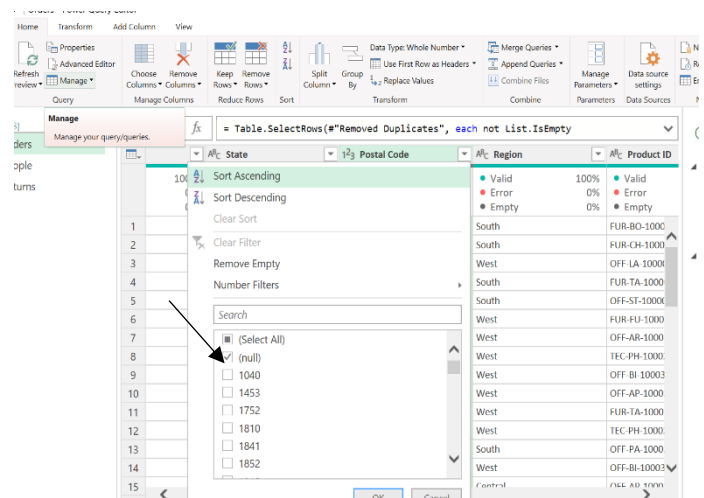


Table: SelectRows("#Removed Duplicates", each not List.IsEmpty)

State	Postal Code	Region
Kentucky		
Kentucky		
California		
Florida		
Florida		
California		
California		
California		
California		
California		
California		

Postal Code statistics: 9983 (99%) Valid, 0 (0%) Error, 11 (< 1%) Empty. Remove Empty button highlighted.

10. replace nulls with postal code of Burlington Vermont after search about it :

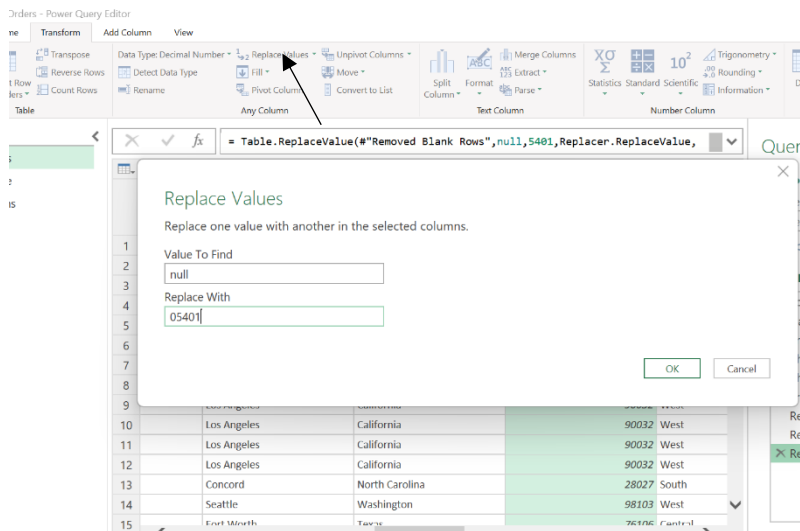


Table: ReplaceValue("#Removed Blank Rows", null, 5401, Replacer.ReplaceValue, ...)

State	Postal Code	Region
Los Angeles		
Los Angeles		
Los Angeles		
Concord		
Seattle		
East North		

Postal Code statistics: 9994 (100%) Valid, 0 (0%) Error, 0 (0%) Empty. Remove Empty button highlighted.

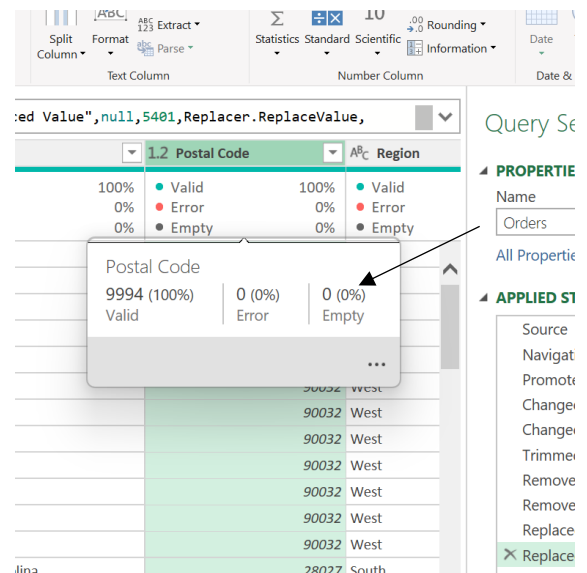


Table: ReplaceValue("#Removed Blank Rows", null, 5401, Replacer.ReplaceValue, ...)

State	Postal Code	Region
Los Angeles		
Los Angeles		
Los Angeles		
Concord		
Seattle		
East North		

Postal Code statistics: 9994 (100%) Valid, 0 (0%) Error, 0 (0%) Empty. Remove Empty button highlighted.

Final result

11.add custom column (profit margin):

1.

The screenshot shows the Power Query Editor with the 'Add Column' tab selected. The formula bar displays: `= Table.AddColumn(#"Replaced Value", "Profit Margin", each [Profit]/[Sales])`. The data table has columns: Quantity, 1.2 Discount, 1.2 Profit, and Profit Margin. The 'Profit Margin' column is highlighted in blue.

	Quantity	1.2 Discount	1.2 Profit	Profit Margin
Valid	100%	Valid	100%	Valid
Error	0%	Error	0%	Error
Empty	0%	Empty	0%	Empty
1	2	0	41.9136	0.16
2	3	0	219.582	0.3
3	2	0	6.8714	0.47
4	5	0.45	-383.031	-0.4
5	2	0.2	2.5164	0.1125
6	7	0	14.1694	0.29
7	4	0	1.9656	0.27
8	6	0.2	90.7152	0.1
9	3	0.2	5.7825	0.3125
10	5	0	34.47	0.3
11	9	0.2	85.3092	0.05
12	4	0.2	68.3568	0.075
13	3	0.2	5.4432	0.35
14	3	0.2	132.5922	0.325
15	5	0.8	122.858	1.8

2.write the formula:

The screenshot shows the 'Custom Column' dialog box in the Power Query Editor. The 'New column name' is 'profit margin'. The 'Custom column formula' is `= [Profit]/[Sales]`. The 'Available columns' list on the right includes: Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit, and Profit Margin. The 'Sales' column is selected. The 'OK' button is highlighted.

Custom Column

Add a column that is computed from the other columns.

New column name
profit margin

Custom column formula
`= [Profit]/[Sales]`

Available columns
Category
Sub-Category
Product Name
Sales
Quantity
Discount
Profit
Profit Margin

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

3.change data type to percentage:

Custom,{"Profit Margin",

Profit Margin

1.2 Decimal Number
\$ Currency
123 Whole Number
% Percentage
Date/Time
Date
Time
Date/Time/Timezone
Duration
Text
True/False
Binary
Using Locale...

Query Settings

PROPERTIES

Name
Orders

All Properties

APPLIED STEPS

Source
Navigation
Promoted Headers
Changed Type
Changed Type1
Trimmed Text
Removed Duplicates
Removed Blank Row
Replaced Value
Added Custom
X Changed Type2

12.add sales category:

1. knowing the statistical analysis of sales:(max,min,avg,std)

Orders - Power Query Editor

File Home Transform Add Column View

Query settings Layout Data Preview

Table.AddColumn("#Replaced

Query Dependencies

View a dependency tree of all queries.

each [Profit]/[Sales]

Quantity

Column statistics

Error 0
Empty 0
Distinct 6144
Unique 4272
NaN 0
Zero 0
Min 0.44399...
Max 22638.48
Average 229.858...
Standard deviation 623.245...

Value distribution

12.96 19.44 15.52 10.368 32.4 25.92 17.94 6.48 20.736 14.94 45.36 10.272 11.52 5.76 14.62 31.104 8.64 25.92

Query Settings

PROPERTIES

Name
Orders

All Properties

APPLIED STEPS

Source
Navigation
Promoted Headers
Changed Type
Changed Type1
Trimmed Text
Removed Duplicates
Removed Blank Rows
Replaced Value
X Added Custom

2. make the sales category:

The screenshot shows the Power Query Editor interface. The 'Add Column' tab is active, and the formula bar contains the following formula:

```
= Table.AddColumn(#"Replaced Value", "Profit Margin", each [Profit]/[Sales])
```

The data table below shows the results of the query, including the new 'Profit Margin' column.

	Quantity	1.2 Discount	1.2 Profit	Profit Margin
Valid	100%	Valid	100%	Valid
Error	0%	Error	0%	Error
Empty	0%	Empty	0%	Empty
1	2	0	41.9136	0.16
2	3	0	219.582	0.3
3	2	0	6.8714	0.47
4	5	0.45	-383.031	-0.4
5	2	0.2	2.5164	0.1125
6	7	0	14.1694	0.29
7	4	0	1.9656	0.27
8	6	0.2	90.7152	0.1
9	3	0.2	5.7825	0.3125
10	5	0	34.47	0.3
11	9	0.2	85.3092	0.05

.write the formula based on the statistical analysis:

The screenshot shows the 'Add Conditional Column' dialog box in the Power Query Editor. The dialog is used to create a new column based on conditional logic. The 'New column name' is 'Sales Category'. The conditions are defined as follows:

Column Name	Operator	Value	Output
If Sales	is greater than or...	1000	VIP
Else If Sales	is greater than or...	250	High
Else If Sales	is greater than or...	50	Medium
Else			low

13.remove duplicates from returns:(product name is unknown)

The screenshot shows the Power Query Editor interface. The ribbon is set to 'Transform', and the 'Remove Duplicates' button is highlighted. The query name is 'Returns'. The data table has two columns: 'Returned' and 'Order ID'. The 'Removed Duplicates' step is listed in the 'APPLIED STEPS' pane on the right.

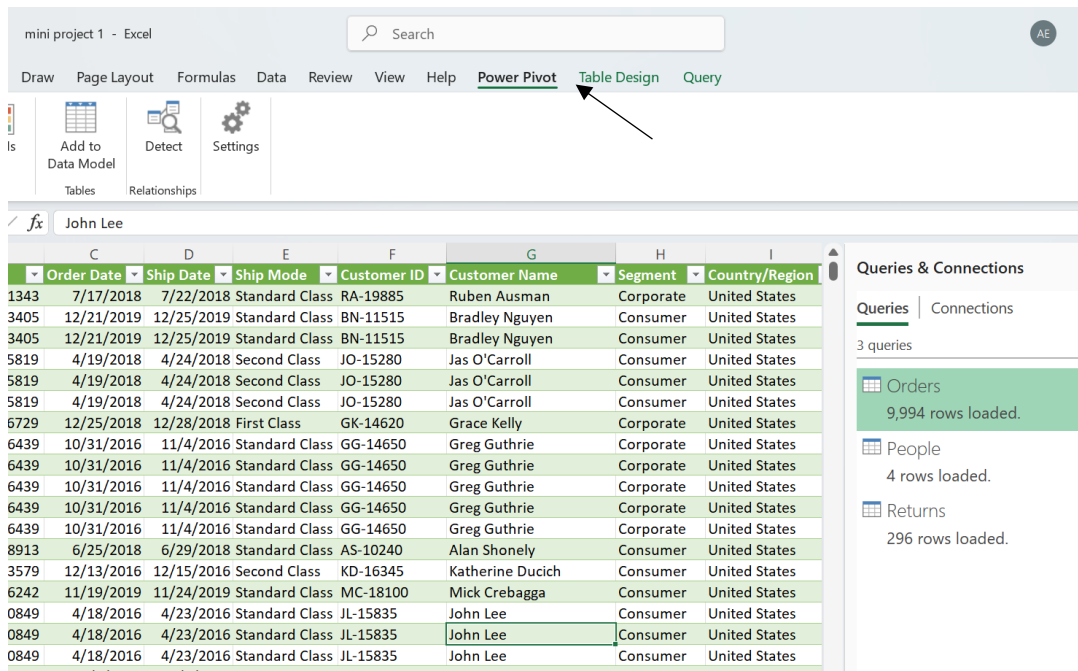
Returned	Order ID
Yes	CA-2016-100762
Yes	CA-2016-100867
Yes	CA-2016-102652
Yes	CA-2016-103373
Yes	CA-2016-103744
Yes	CA-2016-103940
Yes	CA-2016-104829
Yes	CA-2016-105270
Yes	CA-2016-108609
Yes	CA-2016-108861
Yes	CA-2016-109918
Yes	CA-2016-110786
Yes	CA-2016-111871
Yes	CA-2016-116785

14. make the second row as a header:(people also)

The screenshot shows the Power Query Editor interface. The ribbon is set to 'Transform', and the 'Use First Row as Headers' button is highlighted. The query name is 'Returns'. The data table has two columns: 'Returned' and 'Order ID'. The 'Use First Row as Headers' step is listed in the 'APPLIED STEPS' pane on the right.

Returned	Order ID
Yes	CA-2016-100762
Yes	CA-2016-100867
Yes	CA-2016-102652
Yes	CA-2016-103373
Yes	CA-2016-103744
Yes	CA-2016-103940
Yes	CA-2016-104829
Yes	CA-2016-105270
Yes	CA-2016-108609
Yes	CA-2016-108861
Yes	CA-2016-109918
Yes	CA-2016-110786
Yes	CA-2016-111871
Yes	CA-2016-116785

15.open power pivot to make some relations:



mini project 1 - Excel

Search

Draw Page Layout Formulas Data Review View Help **Power Pivot** Table Design Query

Is Add to Data Model Detect Settings

Tables Relationships

John Lee

	C	D	E	F	G	H	I
	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region
1343	7/17/2018	7/22/2018	Standard Class	RA-19885	Ruben Ausman	Corporate	United States
3405	12/21/2019	12/25/2019	Standard Class	BN-11515	Bradley Nguyen	Consumer	United States
3405	12/21/2019	12/25/2019	Standard Class	BN-11515	Bradley Nguyen	Consumer	United States
5819	4/19/2018	4/24/2018	Second Class	JO-15280	Jas O'Carroll	Consumer	United States
5819	4/19/2018	4/24/2018	Second Class	JO-15280	Jas O'Carroll	Consumer	United States
5819	4/19/2018	4/24/2018	Second Class	JO-15280	Jas O'Carroll	Consumer	United States
6729	12/25/2018	12/28/2018	First Class	GK-14620	Grace Kelly	Corporate	United States
6439	10/31/2016	11/4/2016	Standard Class	GG-14650	Greg Guthrie	Corporate	United States
6439	10/31/2016	11/4/2016	Standard Class	GG-14650	Greg Guthrie	Corporate	United States
6439	10/31/2016	11/4/2016	Standard Class	GG-14650	Greg Guthrie	Corporate	United States
6439	10/31/2016	11/4/2016	Standard Class	GG-14650	Greg Guthrie	Corporate	United States
6439	10/31/2016	11/4/2016	Standard Class	GG-14650	Greg Guthrie	Corporate	United States
8913	6/25/2018	6/29/2018	Standard Class	AS-10240	Alan Shonely	Consumer	United States
3579	12/13/2016	12/15/2016	Second Class	KD-16345	Katherine Ducich	Consumer	United States
6242	11/19/2019	11/24/2019	Standard Class	MC-18100	Mick Crebaggga	Consumer	United States
0849	4/18/2016	4/23/2016	Standard Class	JL-15835	John Lee	Consumer	United States
0849	4/18/2016	4/23/2016	Standard Class	JL-15835	John Lee	Consumer	United States
0849	4/18/2016	4/23/2016	Standard Class	JL-15835	John Lee	Consumer	United States

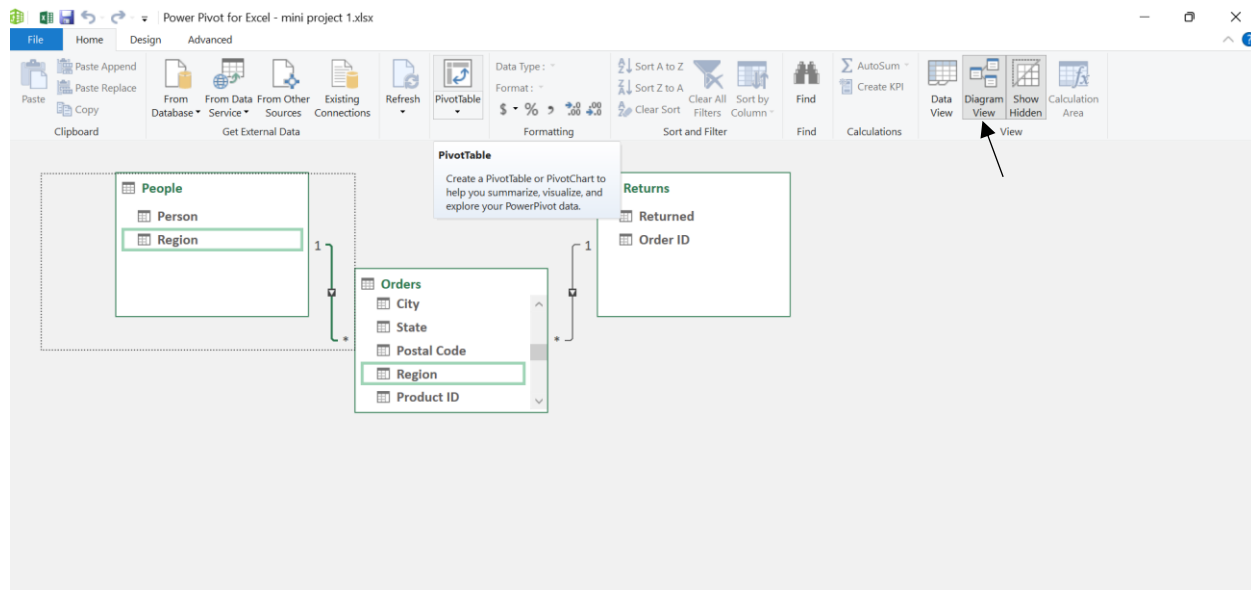
Queries & Connections

Queries | Connections

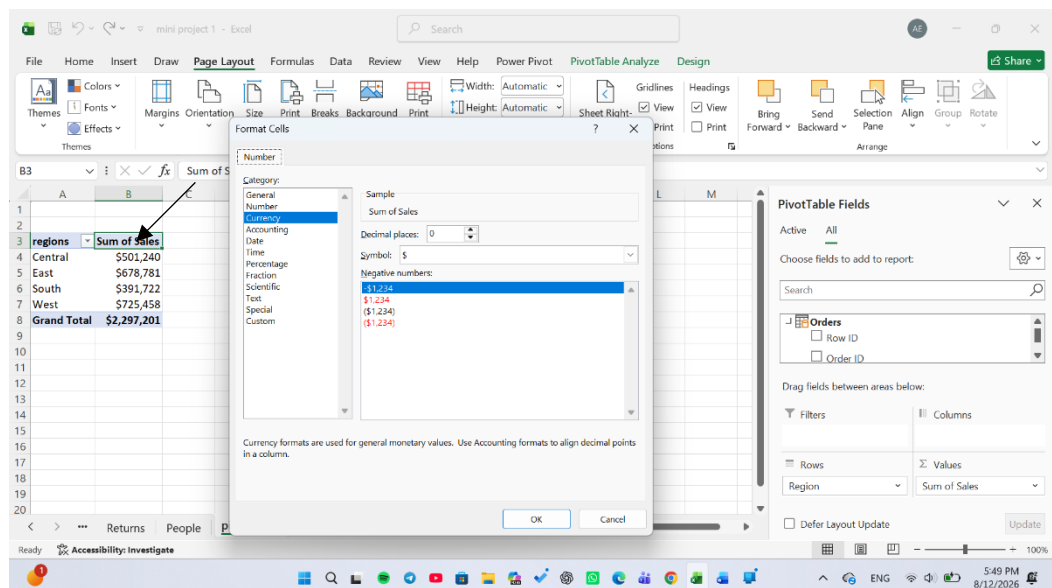
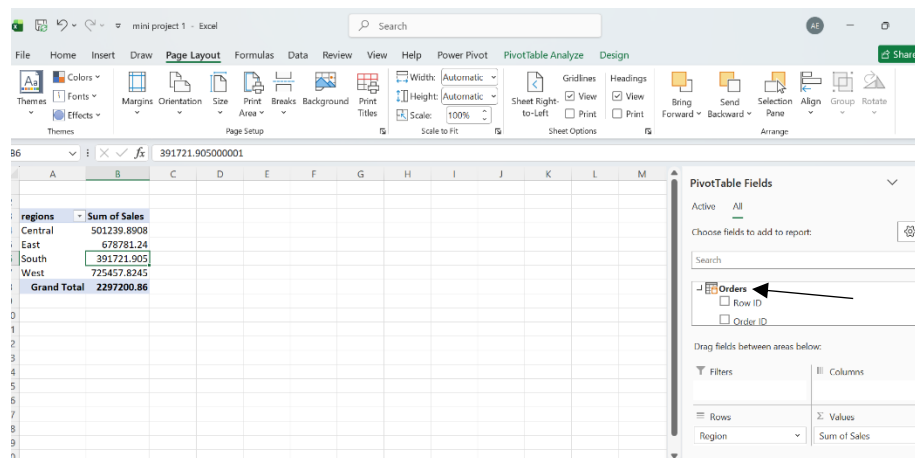
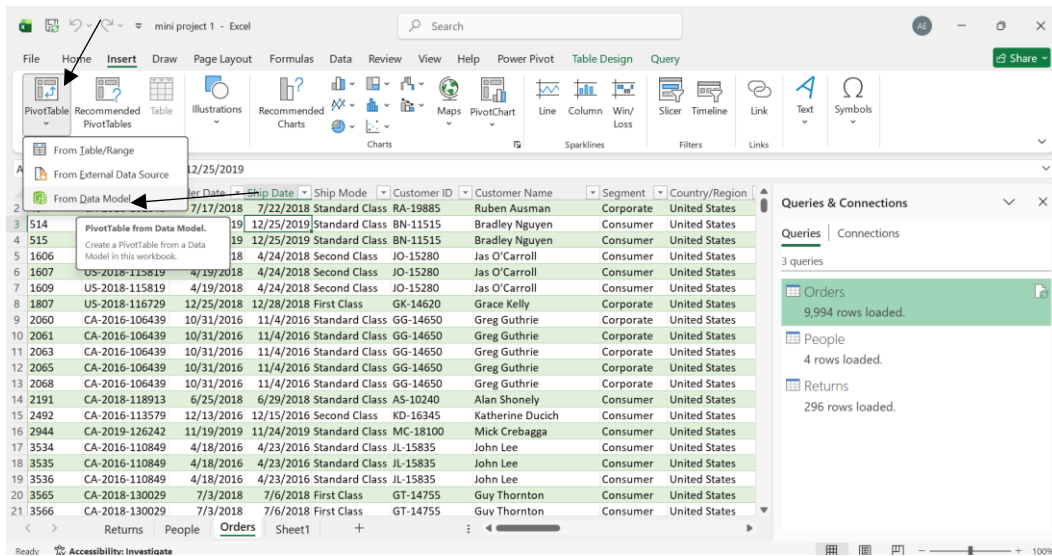
3 queries

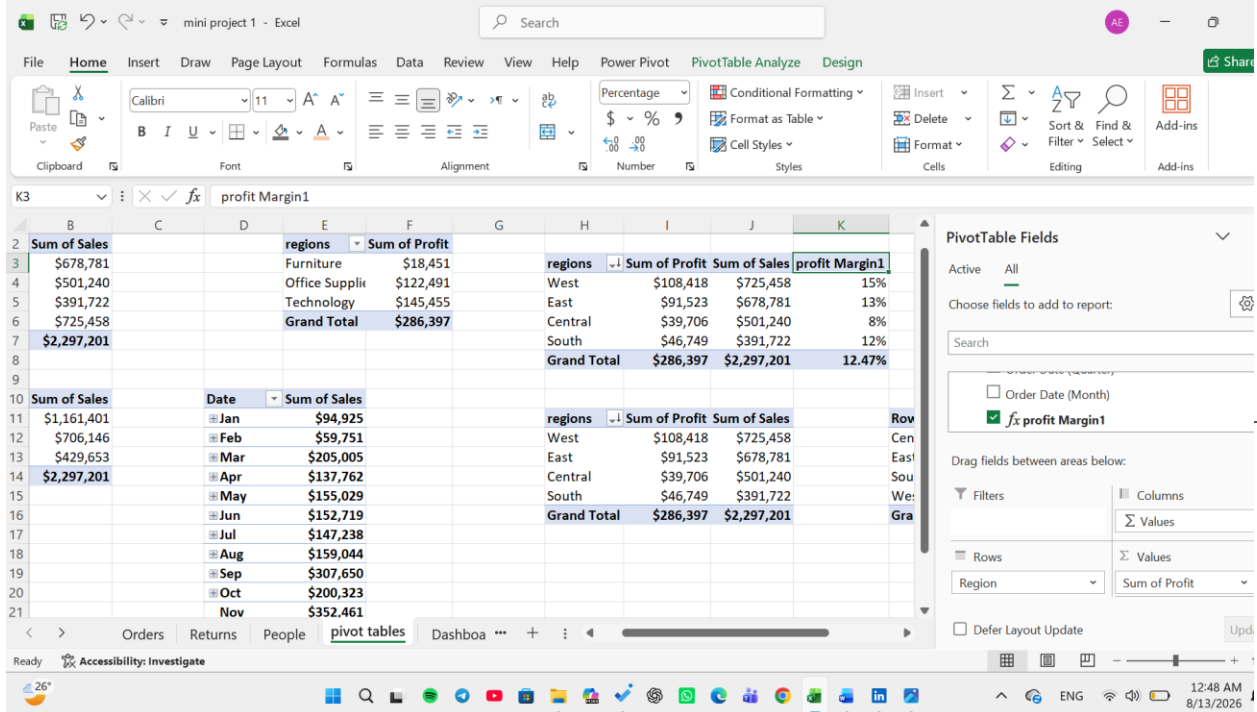
- Orders
9,994 rows loaded.
- People
4 rows loaded.
- Returns
296 rows loaded.

drag and drop:(order ID, region)(one to many)



16.insert pivot tables:





dashboard:

